y →

x ↓

	0	P	R	O	V	I	D	E	N	C	E
0	0	0	0	0	0	0	0	0	0	0	0
P	0	↖ 1	← 1	← 1	← 1	← 1	← 1	← 1	← 1	← 1	← 1
R	0	1 ↑	↖ 2	← 2	← 2	← 2	← 2	← 2	← 2	← 2	← 2
E	0	1 ↑	2 ↑	← 2	← 2	← 2	← 2	↖ 3	← 3	← 3	↖ 3
S	0	1 ↑	2 ↑	← 2	← 2	← 2	← 2	3 ↑	← 3	← 3	← 3
I	0	1 ↑	2 ↑	← 2	← 2	↖ 3	← 3	← 3	← 3	← 3	← 3
D	0	1 ↑	2 ↑	← 2	← 2	↑ 3	↖ 4	← 4	← 4	← 4	← 4
E	0	1 ↑	2 ↑	← 2	← 2	3 ↑	4 ↑	↖ 5	← 5	← 5	↖ 5
N	0	1 ↑	2 ↑	← 2	← 2	3 ↑	4 ↑	5 ↑	↖ 6	← 6	← 6
T	0	1 ↑	2 ↑	← 2	← 2	3 ↑	4 ↑	5 ↑	6 ↑	← 6	← $\boxed{6}$

	0	
0	0	0
P	0	↖ 1

	0	R
0	0	0
P	0	← 0 ↑

P R I D E N

= 6

⇒ if (x[i] = = y[j])

{ c[i][j] = c[i-1] [j-1] + 1

X = A B C B D A B

Y = B D C A B A

	0	B	D	C	A	B	A
0	0	0	0	0	0	0	0
A	0	↑ 0	↑ 0	↑ 0	↖ 1	← 1	↖ 1
B	0	↖ 1	← 1	← 1	1 ↑	↖ 2	← 2
C	0	1 ↑	1 ↑	↖ 2	← 2	2 ↑	2 ↑
B	0	↖ 1	1 ↑	2 ↑	2 ↑	↖ 3	← 3
D	0	1 ↑	↖ 2	2 ↑	2 ↑	3 ↑	3 ↑
A	0	1 ↑	2 ↑	2 ↑	↖ 3	3 ↑	↖ 4
B	0	↖ 1	2 ↑	2 ↑	3 ↑	↖ 4	4 ↑

B C B A

$X = A\ B\ C\ B\ D\ A\ B$

$Y = B\ D\ C\ A\ B\ A$

	0	B	D	C	A	B	A
0	0	0	0	0	0	0	0
A	0	↑ 0	↑ 0	↑ 0	↖ 1	← 1	↖ 1
B	0	↖ 1	← 1	← 1	1 ↑	↖ 2	← 2
C	0	1 ↑	1 ↑	↖ 2	← 2	2 ↑	2 ↑
B	0	↖ 1	1 ↑	2 ↑	2 ↑	↖ 3	← 3
D	0	1 ↑	↖ 2	2 ↑	2 ↑	3 ↑	3 ↑
A	0	1 ↑	2 ↑	2 ↑	↖ 3	3 ↑	↖ 4
B	0	↖ 1	2 ↑	2 ↑	3 ↑	↖ 4	← 4

B C B A

B C A B

X = A B C B D A B

Y = B D C A B A

	0	B	D	C	A	B	A
0	0	0	0	0	0	0	0
A	0	↑0	↑0	↑0	↖1	←1	↖1
B	0	↖1	←1	←1	↑1	↖2	←2
C	0	↑1	↑1	↖2	←2	↑2	↑2
B	0	↖1	↑1	↑2	↑2	↖3	←3
D	0	↑1	↖2	↑2	↑2	↑3	↑3
A	0	↑1	↑2	↑2	↖3	↑3	↖4
B	0	↖1	↑2	↑2	↑3	↖4	←4

$$\left.\begin{array}{l} B\ C\ B\ A \\ B\ C\ A\ B \end{array}\right\} 4$$

(i)

1	P	0	↖ 1	← 1	← 1	← 1	← 1	← 1	← 1	← 1	← 1	← 1
2	R	0	1 ↑	↖ 2	← 2	← 2	← 2	← 2	← 2	← 2	← 2	← 2
3	E	0	1 ↑	2 ↑	← 2	← 2	← 2	← 2	↖ 3	← 3	← 3	↖ 3
4	S	0	1 ↑	2 ↑	← 2	← 2	← 2	← 2	3 ↑	← 3	← 3	← 3
5	I	0	1 ↑	2 ↑	← 2	← 2	↖ 3	← 3	← 3	← 3	← 3	← 3
6	D	0	1 ↑	2 ↑	← 2	← 2	↑ 3	↖ 4	← 4	← 4	← 4	← 4
7	E	0	1 ↑	2 ↑	← 2	← 2	3 ↑	4 ↑	↖ 5	← 5	← 5	↖ 5
8	N	0	1 ↑	2 ↑	← 2	← 2	3 ↑	4 ↑	5 ↑	↖ 6	← 6	← 6
9	T	0	1 ↑	2 ↑	← 2	← 2	3 ↑	4 ↑	5 ↑	6 ↑	← 6	$\boxed{6}$

		0
	0	0
	P	0

P R I D E N

6

⇒ If (x[i] = = y[j])

{ c[i][j] = c[i-1][j-1] + 1 }

else { c[i][j] = max(c[i][j-1], c[i-1][j]) }